THE EQUIVALENCE BETWEEN BOUNDED ABOVE COHERENT AND COHERENT-IN-COHOMOLOGY BOUNDED ABOVE DERIVED CATEGORIES ON NOETHERIAN SCHEMES

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Allowed standard prerequisites (editorial).

- Coherent sheaves form a Serre subcategory of quasi-coherent sheaves on a Noetherian scheme. invoked at proof lines 2793-2799. One of the two conditions of the derived category embedding lemma.
- Every quasi-coherent module surjecting onto a coherent module has a coherent submodule still surjecting onto it, via the filtered union of coherent submodules. established at proof lines 2800-2807. The second condition of the derived category embedding lemma (resolution property of the inclusion).
- The derived category criterion for a Serre subcategory inclusion to induce a fully faithful and essentially surjective embedding on bounded above derived categories. invoked at proof lines 2808-2809. Concludes the equivalence.

Source reference. Original definition 02MO: sources/stacks-chow/tags.txt, source lines 3451–3451. Cited at proof lines 2794-2795 in the Serre subcategory observation; the verification and conclusion are inside the counted span.

Source reference. Original lemma 02MP: sources/stacks-chow/tags.txt, source lines 3452–3452. Cited at proof lines 2796-2796 in the Serre subcategory observation; the verification and conclusion are inside the counted span.

Source reference. Original lemma 01Y0: sources/stacks-chow/tags.txt, source lines 2539–2539. Cited at proof lines 2797-2798 in the Serre subcategory observation; the verification and conclusion are inside the counted span.

Source reference. Original lemma 01Y1: sources/stacks-chow/tags.txt, source lines 2540–2540. Cited at proof lines 2798-2799 in the Serre subcategory observation; the verification and conclusion are inside the counted span.

Source reference. Original lemma 01PG: sources/stacks-chow/tags.txt, source lines 2229–2229. Cited at proof lines 2805-2806 for the filtered union of coherent submodules; the lifting argument and conclusion are inside the counted span.

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Source reference. Original lemma 0FCL: sources/stacks-chow/tags.txt, source lines 19186–19186. Cited at proof lines 2808-2809 to conclude once its two conditions are verified inside the counted span.

Lemma 0.2. *Let X be a Noetherian scheme. Then the functor*

$$D^-(Coh(\mathcal{O}_X)) \longrightarrow D^-_{Coh(\mathcal{O}_X)}(QCoh(\mathcal{O}_X))$$

is an equivalence.

Proof. Observe that $Coh(\mathcal{O}_X) \subset QCoh(\mathcal{O}_X)$ is a Serre subcategory, see Homology, Definition 02MO and Lemma 02MP and Cohomology of Schemes, Lemmas 01Y0 and 01Y1. On the other hand, if $\mathcal{G} \to \mathcal{F}$ is a surjection from a quasi-coherent $\mathcal{O}_X$-module to a coherent $\mathcal{O}_X$-module, then there exists a coherent submodule $\mathcal{G}' \subset \mathcal{G}$ which surjects onto $\mathcal{F}$. Namely, we can write $\mathcal{G}$ as the filtered union of its coherent submodules by Properties, Lemma 01PG and then one of these will do the job. Thus the lemma follows from Derived Categories, Lemma 0FCL. □